

ARSLAN MAJAL

Machine Learning Researcher | Robust Estimation and Uncertainty-Aware Learning

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U.S. Work Authorization: STEM OPT through July 2028 - No Current Sponsorship Required

PROFESSIONAL PROFILE

Research Engineer with graduate training in machine learning, applied mathematics, Bayesian inference, and statistical modeling. Develops reliable estimation and learning methods for systems affected by outliers, biased measurements, heteroscedastic noise, and imperfect data. Experienced with Python and MATLAB implementation, simulation design, quantitative benchmarking, uncertainty analysis, and academic writing, with applications in robotics, perception, localization, GNSS reliability, image processing, and sensor fusion.

RESEARCH INTERESTS

Robust Gaussian Process Regression	Outlier-Robust Neural Network Training	Uncertainty-Aware and Adaptive Bayesian Filtering
Machine Learning under Heteroscedastic Noise	Sensor Fusion and State Estimation	Uncertainty Quantification for Perception and Control

EDUCATION

M.S. in Mechanical Engineering - GPA: 3.83

Sep 2023 - May 2025

University of Wisconsin-Madison

Thesis: "Quantifying Estimation Errors in Robust Vision-Based Control: An Extreme Value Theory Approach." Relevant study included artificial neural networks, matrix methods for machine learning, computer vision, image processing, optimization, probability, and statistics.

B.S. in Computer Engineering - GPA: 3.83

Sep 2016 - Aug 2020

University of Engineering and Technology, Lahore

Thesis: "Received Signal Strength (RSS)-Based Neural Network for Localization." Graduated as Gold Medalist for best academic performance.

PROFESSIONAL AND RESEARCH EXPERIENCE

Research Engineer

Jul 2026 - Present

Zunitech LLC

- Conduct research and development in robust machine learning, Bayesian state estimation, and uncertainty-aware inference.
- Develop and evaluate Python and MATLAB algorithms for detecting and mitigating corrupted measurements, with applications in robotics, GNSS reliability, image processing, and sensor-based systems.

Graduate Researcher

Sep 2023 - May 2025

ARC Lab, University of Wisconsin-Madison

- Applied Extreme Value Theory to estimate local norm error bounds for learning-based and model-based perception algorithms near nominal control trajectories.
- Built Python and MATLAB pipelines for simulation, numerical analysis, uncertainty evaluation, and technical reporting in robust vision-based control.

Research Assistant / Graduate Researcher

Sep 2020 - Aug 2023

Smart Data Systems and Applications Laboratory, LUMS

- Developed variational Bayesian filtering and smoothing algorithms for nonlinear state estimation with outliers, measurement bias, and uncertain noise statistics.
- Led development and evaluation of selective outlier-rejection smoothers for multi-sensor trajectory reconstruction, preserving valid measurement dimensions while suppressing corrupted data.
- Contributed to deep-unfolding methods for low-rank decomposition and multi-degradation image restoration under noise, blur, and rain artifacts.

TEACHING EXPERIENCE

Teaching Assistant - ME 340: Dynamic Systems University of Wisconsin-Madison

Fall 2023 - Spring 2025

- Served across four semesters for a course enrolling more than 150 students; led weekly problem-solving discussion sections, held office hours, graded assignments and examinations, and provided detailed technical feedback.

Mathematics Teacher and AI Advisor Springfield Commonwealth Academy

Sep 2025 - Jun 2026

- Taught grades 6-12 across Pre-Algebra, Algebra I, Algebra II, Geometry, and Precalculus; designed lessons, assessments, and differentiated learning activities.
- Helped the school explore practical and responsible uses of artificial intelligence in teaching and academic workflows.

Geometry Teacher Vertex Charter School

May 2026 - Aug 2026

- Worked with Vertex from May through August, including an intensive Geometry summer term from July through August; developed objectives, lessons, practice packets, quizzes, tests, study materials, and final examinations.
- Provided individualized one-on-one instruction to students with Individualized Education Programs (IEPs), adapting explanations, pacing, and practice to their learning needs.

PUBLICATIONS AND PREPRINTS

- **Dictionary-Based Contrastive Learning for GNSS Jamming Detection.** Z. Hussain, **A. Majal**, A. H. Chughtai, and T. Nadeem. arXiv:2512.07512, 2025. [arXiv]
- **EMORF-II: Adaptive EM-Based Outlier-Robust Filtering with Correlated Measurement Noise.** **A. Majal**, A. H. Chughtai, and M. Tahir. IEEE MLSP, 2025. [IEEE] [arXiv]
- **Trajectory Reconstruction through a Gaussian Adaptive Selective Outlier Rejecting Smoother.** **A. Majal** and A. H. Chughtai. arXiv:2410.20411, 2024. [arXiv]
- **Outlier-Robust Unscented RTS Smoothing for Independent Sensing Data.** **A. Majal** and A. H. Chughtai. IEEE Sensors Letters, vol. 8, no. 10, 2024. [IEEE]
- **Variational-Based Nonlinear Bayesian Filtering with Biased Observations.** A. H. Chughtai, **A. Majal**, M. Tahir, and M. Uppal. IEEE Transactions on Signal Processing, vol. 70, pp. 5295-5307, 2022. [IEEE]

SELECTED TECHNICAL PROJECTS

Deep Learning for Robust Kalman Filtering

Developing a learning-augmented Extended Kalman Filter for outliers, sensor bias, and uncertain noise, with evaluation against classical EKF and learning-based filtering frameworks.

Robust Gaussian Process Regression with Adaptive Noise Learning

Integrating variational Bayesian and Expectation-Maximization updates to learn nominal noise and outlier behavior directly from corrupted regression data.

Multi-Degradation Image Restoration via Deep Unfolding

Combining Orthogonal Variational PCA, content-aware partitioning, adaptive rank estimation, and attention-based refinement for unknown noise, blur, and rain degradations.

Outlier-Robust Neural Network Training

Developing adaptive sample-weighting and robust cross-validation strategies for label noise and unknown data contamination.

TECHNICAL SKILLS

Programming	Python, MATLAB, C++, SQL
Machine Learning	PyTorch, scikit-learn, NumPy, pandas, neural networks, Gaussian processes, regression, clustering
Inference and Estimation	Variational Bayes, Expectation-Maximization, nonlinear Kalman filtering, RTS smoothing, uncertainty quantification
Tools	Jupyter, LaTeX, Git, Excel, Google Sheets, simulation-based analysis, technical writing

HONORS AND AWARDS

- Fulbright Scholarship Selectee, 2021
- Gold Medalist for Best Academic Performance, UET Lahore, 2020